

Compact Validation Tables

Validation Tables

Table 1: Search space and selected settings on the validation set.

Param	Candidates	A1	A2	Note
Grid size ($N \times N$)	{8, 16, 32}	16×16	16×16	Good tradeoff between spatial detail and runtime.
Motion thresh. (%)	{1.0, 1.5, 2.0, 5.0}	2.0	1.5	Lower threshold helps capture subtle smoke motion.
Fire color rule	tuned range	N/A	used	Helps detect bright flame-like regions.
Smoke grayness	candidate intervals	N/A	[0.7, 0.95]	Captures low-saturation semi-transparent smoke.
Temporal window	{3, 5} / minimal	minimal	5	Improves stability and suppresses noise.

Table 2: Search space and selected settings on the validation set.

Group	Parameter	A1	A2	Note
Shared	Grid size ($N \times N$)	16×16	16×16	Good tradeoff between detail and runtime.
Shared	Motion threshold (%)	2.0	1.5	Lower threshold helps capture subtle smoke motion.
Fire color rule	Red threshold	N/A	selected	Detects bright flame-like red regions.
	Saturation threshold	N/A	selected	Helps reject non-fire bright objects.
	Intensity threshold	N/A	selected	Keeps flame response focused on strong regions.
Smoke grayness rule	Grayness range	N/A	[0.7, 0.95]	Captures low-saturation smoke appearance.
	Texture consistency	N/A	selected	Helps distinguish smoke from random motion.
Temporal rule	Window size	minimal	5	Improves stability and suppresses noise.

Table 3: Validation results under the Recall(Anomaly) $\geq 99\%$ safety constraint.

Config	Rec.	Filt.	Score	Safe?	Role
A1: 16×16, 2.0%, motion only	97.2%	65.0%	N/A	No	Motion-only baseline; fails safety target.
A2: 16×16, 1.5%, smoke rules	99.1%	72.1%	0.714	Yes	Selected final configuration.
A2: 16×16, 2.0%, smoke rules	99.0%	68.3%	0.676	Yes	Safe, but less efficient.
A2: 32×32, 1.0%, smoke rules	99.2%	65.7%	0.652	Yes	Slightly higher recall, lower efficiency.

Note: Approach 1 (A1) is included as a simple motion-only baseline. Its best configuration remains below the 99% recall target, so it is not selected for final deployment.